Workshop : How to incorporate AI into your Research process

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1 Define your research problem

- To formulate a research problem within your field of study, consider leveraging AI chatbots. Begin by accessing an AI chatbot such as ChatGPT, Google BARD, or Perplexity.
- For instance, you can ask chatGPT or google BARD the following question (replace [*my field of study*] with your field) :

  "**Identify a specific area of interest within [*my field of study*] that you believe warrants further investigation. What issues, gaps, or questions do you observe in this area ?**".
- If you aim to address a real-world and practical problem, consider posing the following question:

  "**Think about the practical implications of [*my field of study*]. Are there real-world problems or issues that require investigation to develop innovative solutions or informed strategies ?**"
- If you are utilizing an internet-connected chatbot such as Google BARD or Perplexity, it is advisable to pose the following question:

  "**Could you review the academic literature in [*my field of study*] from the past five years? What questions have these studies left open for future research?**".
- Now identify a specific problem and continue with the following steps. In the example given, we picked the area of **quantitative economics**, and there are some open research questions. We'll focus on the first one.

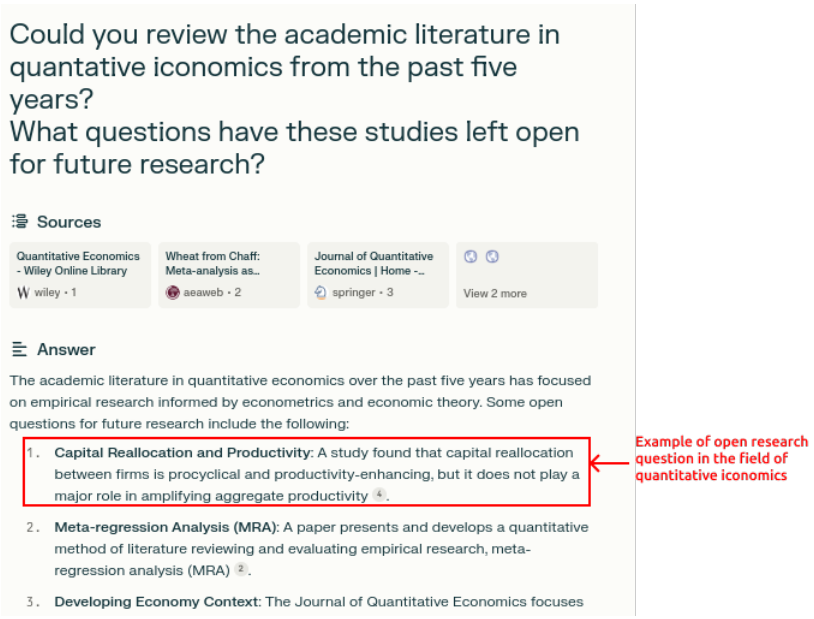


Figure 1: Identifying a research problem

2 Specific research problem

Once you defined a research problem, you could make a more specific question like :

"**Come up with creative research angles to study** ***[the selected problem]*** **?**"



Figure 2: Specific research problem

3 List research questions

Now that you've identified the specific research problem you want to address, you can ask the chatbot to generate and list some research questions. For instance : "**List ten potential research**

questions related to [the selected problem]".

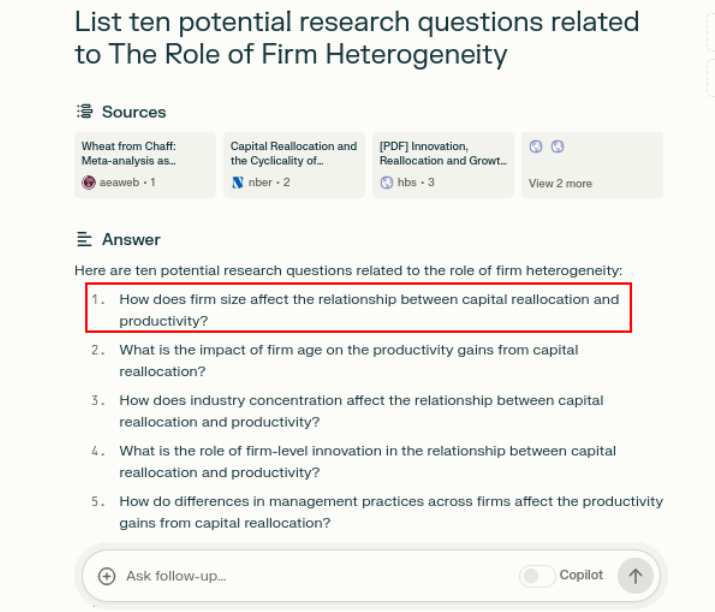


Figure 3: List research questions

4 Develop an outline from scratch

Once we have selected one or more research questions, it's time to develop an outline for our academic work. You can ask the AI chatbot the following question :

Could you help me draft a preliminary outline for my academic paper on [*the selected question*] ? I'd appreciate insights on how to arrange the introduction, literature review, methodology, results, and conclusion to ensure a cohesive and well-flowing document. Outline the main sections, subtopics, and key points to be covered in the paper.

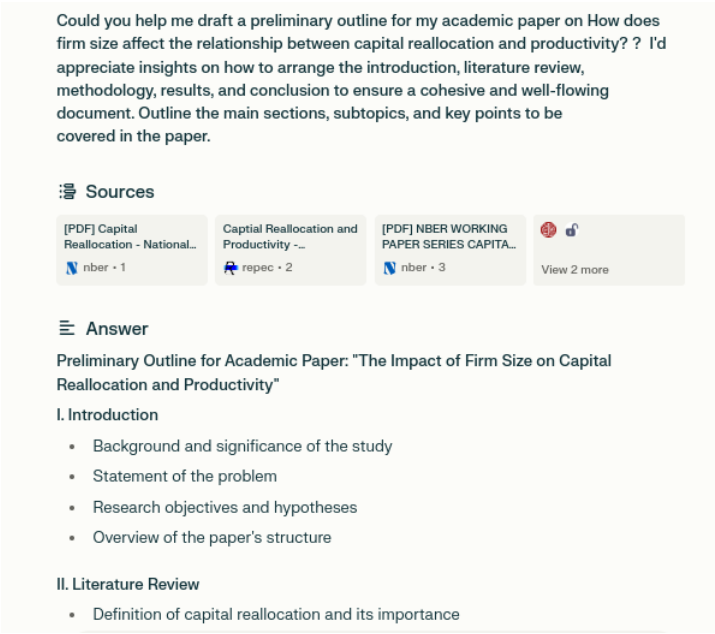


Figure 4: Develop an outline

5 Detail the outline

Once you have the outline, you can delve deeper and expand on it by asking the AI chatbot to elaborate on each section of your outline. This can be done as you continue to make advancements in your study.

- For the introductory section, we can use :

  Develop the introductory section of the paper, including the abstract, background information, and research objectives.

- For the literature review section we can ask :

  Outline the literature review section, identifying key sources and organizing them based on themes, theories, or chronological relevance.

- **Draft a section discussing the implications of my findings in relation to existing literature and their significance in the field.**

- For the conclusion :

  Outline the conclusion, summarizing the key points discussed, reiterating the significance of my findings, and possibly suggesting practical applications.

With this, you have crafted a comprehensive outline for your work, and you can now proceed to the next steps.

6 Conduct a literature review

Please don't use ChatGPT for this task, as GPT may not perform well in conducting literature reviews.

For this purpose, I recommend utilizing Perplexity. You can ask :
Identify studies published within the last five years that examine [*the selected subject*].

- It will provide you with a concise list of articles. I suggest utilizing Google Scholar, Connected Papers, and Elicit to delve into further research papers.

- Now, it's time to establish a **screening strategy**: determine criteria for filtering some articles while retaining others. To achieve this, you can employ the following prompt:

  Develop a screening strategy to determine the relevance of gathered sources for my research. What criteria should I consider to evaluate their suitability ?

- Feel free to employ ChatDOC or ChatPDF to filter your articles, retaining only those essential for your literature review. Subsequently, summarize the pertinent articles, extracting and preserving crucial information.

7 Develop a methodology

This stage varies significantly based on each project's requirements, but a broad prompt you might pose is:

Create a research methodology for exploring *[the subject]*.

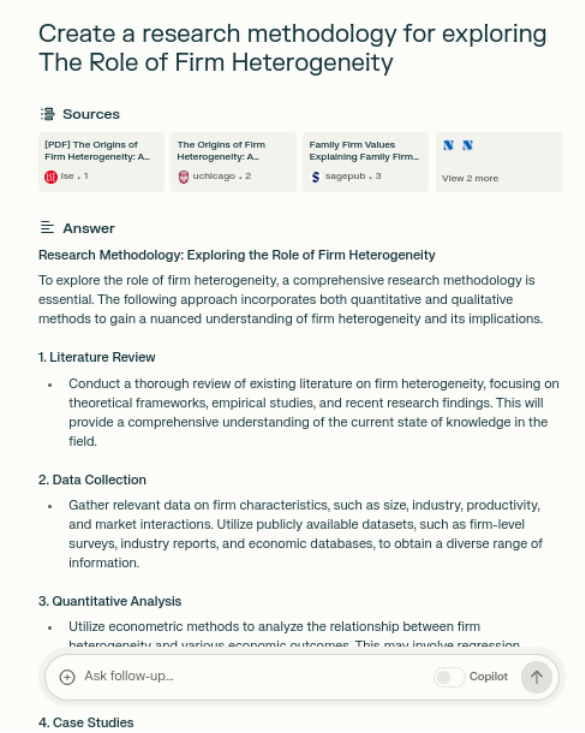


Figure 5: Develop a methodology